Quantum Hardware Design and Optimization
Laboratory 1 – Exercise 1

Group 4

Evaluation by hand of the state evolution

Two-qubit states are written as $|q_0 q_1\rangle$, with the basis ordered as $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Layer 0 – initial state

$$|\psi_{t_0}\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\\0\\0\end{pmatrix} \rightarrow |00\rangle .$$

Layer 1 – $H \otimes H$

$$G_1 = H \otimes H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix} \otimes \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & 1 & 1 & 1\\1 & -1 & 1 & -1\\1 & 1 & -1 & -1\\1 & -1 & -1 & 1\end{pmatrix},$$

$$|\psi_{t_1}\rangle = G_1 |\psi_{t_0}\rangle = \frac{1}{2}\begin{pmatrix}1 & 1 & 1 & 1\\1 & -1 & 1 & -1\\1 & 1 & -1 & -1\\1 & -1 & -1 & 1\end{pmatrix}\begin{pmatrix}1\\0\\0\\0\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix} \rightarrow |++\rangle .$$

Layer 2 – CZ

$$G_2 = CZ = \begin{pmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 1 & 0\\0 & 0 & 0 & -1\end{pmatrix},$$

$$|\psi_{t_2}\rangle = G_2 |\psi_{t_1}\rangle = \frac{1}{2}\begin{pmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 1 & 0\\0 & 0 & 0 & -1\end{pmatrix}\begin{pmatrix}1\\1\\1\\1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1\\1\\1\\-1\end{pmatrix} = \frac{1}{2}\big(|00\rangle + |01\rangle + |10\rangle - |11\rangle\big).$$

Layer 3 – $H \otimes H$

$$|\psi_{t_3}\rangle = (H\otimes H)|\psi_{t_2}\rangle = \frac{1}{4}\begin{pmatrix}1 & 1 & 1 & 1\\1 & -1 & 1 & -1\\1 & 1 & -1 & -1\\1 & -1 & -1 & 1\end{pmatrix}\begin{pmatrix}1\\1\\1\\-1\end{pmatrix} = \frac{1}{4}\begin{pmatrix}2\\2\\2\\-2\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1\\1\\1\\-1\end{pmatrix} = \frac{1}{2}\big(|00\rangle+|01\rangle+|10\rangle-|11\rangle\big).$$

Layer 4 – $X \otimes X$

$$X \otimes X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

$$|\psi_{t_4}\rangle = (X \otimes X)\,|\psi_{t_3}\rangle = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2}\big(-|00\rangle + |01\rangle + |10\rangle + |11\rangle \big).$$

Layer 5 – CZ

The controlled-Z gate can be written as a sum of projectors on the control qubit:

$$CZ = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes Z = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

$$|\psi_{t_5}\rangle = CZ\ |\psi_{t_4}\rangle = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2}\big(-|00\rangle + |01\rangle + |10\rangle - |11\rangle \big).$$

Layer 6 – $X \otimes X$

$$|\psi_{t_6}\rangle = (X \otimes X)\,|\psi_{t_5}\rangle = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2}\big(-|00\rangle + |01\rangle + |10\rangle - |11\rangle \big).$$

Layer 7 – $H \otimes H$

$$|\psi_{t_7}\rangle = (H \otimes H)\,|\psi_{t_6}\rangle = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 0 \\ 0 \\ 0 \\ -4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} = -|11\rangle\,.$$

Conclusion

The final state is $-|11\rangle$, i.e. $|11\rangle$ up to a global phase, so a measurement returns 11 with probability 1.

The circuit is one iteration of Grover's algorithm on two qubits:

- layer 1 creates the uniform superposition $|{+}{+}\rangle$;
- layer 2 (CZ) is the *oracle*, which flips the phase of the marked state $|11\rangle$;
- layers 3–7 ($H^{\otimes 2}\,X^{\otimes 2}\,CZ\,X^{\otimes 2}\,H^{\otimes 2}$) form the *diffuser* (inversion about the mean), which amplifies the amplitude of the marked state.

For $N = 4$ elements a single Grover iteration is enough to find the marked element with certainty.